

Quantum Technology: The World Of Very Small Things [C1]

Per molti, la fisica quantistica è un mistero, qualcosa di difficile comprensione. Eppure molte delle tecnologie che utilizziamo ogni giorno si basano proprio su questa scienza.

Quantum' is a word we hear with increasing frequency in science, technology and even business conversations. But what does it actually mean? In essence, 'quantum' refers to the strange and fascinating world of very small things: the [behaviour](#) of atoms, electrons and subatomic particles that form the [building blocks](#) of everything around us.

DIFFERENT LAWS

At the microscopic level, the rules of physics are completely different from those that govern larger objects in everyday life. In quantum physics, particles can exist in two places at once, behave like [waves](#), and even change when we observe them. All of this may sound counterintuitive, but for more than one hundred years, scientists have known that the universe is full of these surprising phenomena. Even if quantum physics seems complicated, most of us already use technologies made possible by quantum science: the [MRI machines](#) in hospitals, the chips inside our phones, even the lasers in supermarket scanners or barcode readers. None of these would exist without quantum physics.

BIG POSSIBILITIES

In the past, controlling quantum [behaviour](#) meant using enormous machines like particle accelerators or complex laboratory instruments to observe large groups of particles. Today, thanks to advances in laser technology, superconducting circuits, and photonic systems, scientists can isolate and manipulate individual atoms or photons with great precision. This ability to control single particles is opening the door to a new

generation of technologies, such as quantum computers, sensors, and secure communication. The best-known of these innovations is quantum computing. While classical computers work with bits, 0 or 1, quantum computers use [qubits](#), which can be 0 and 1 at the same time. This makes them very powerful and a fast way to find solutions for certain problems.

Places

The Rise of the Machines: National Museum of Computing I computer fanno parte della nostra vita quotidiana, per questo ci dimentichiamo che in realtà sono una tecnologia relativamente recente. Un museo nel sud dell'Inghilterra raccoglie la loro storia dalla Seconda guerra mondiale fino ad oggi. Conor Gleeson , Sarah Davison

SUPERPOSITION

To learn more about the basics of quantum science and its real-world applications, Speak Up met with Daniel Goldsmith, Senior Quantum Computing Technologist at Digital Catapult, the London-based innovation agency that works with startups, investors, the public sector, and academia to accelerate the adoption of new and emerging technologies. Goldsmith began by explaining the origin of the word 'quantum' and why [qubits](#) are so extraordinary when compared to classical bits. He introduced the concept of superposition, a key quantum property that allows [qubits](#) to hold much more information than traditional bits. **Daniel Goldsmith (English accent):** The word 'quantum' is from the Latin, and it means an 'amount', a '[lump](#)'. So the idea of quantum mechanics is that energy comes in [lumps](#) or in quantum. The [qubits](#) can be in a superposition of two states at the same time. So the qubit can literally be both 0 and 1 at the same time. And it's only when you observe it that on a probabilistic basis, it [splits](#) itself and it [shows its hand](#) as either 0 or 1. And this means that the [qubits](#) can hold much more data than the classical bits. And that's one of the reasons that quantum computing has got the potential to be much more powerful than classical computing. Google CEO Sundar Pichai, who began his career as a materials engineer.

RICHARD FEYNMAN

Researchers believe that future quantum computers could solve problems that even the best supercomputers cannot manage. They may help design new materials, improve medicines, simulate nature and make [smarter](#) decisions in areas like transport, energy or finance. Goldsmith credited American physicist Richard Feynman for the idea of quantum computers.

Daniel Goldsmith: Quantum computers were invented by Richard Feynman because he realised that simulating quantum systems on a classical computer was a problem that increased in complexity exponentially. So he suggested if you need to understand a quantum system, then you need a quantum computer to best understand it.

UNIMAGINABLE

But quantum technology is much more than really powerful computers. Quantum sensors can measure extremely small changes in gravity or magnetic fields, helping engineers discover cracks inside materials or allowing scientists to detect hidden structures underground. Quantum communication could make our digital networks safer, because quantum signals cannot be intercepted without leaving a [trace](#). And quantum navigation may one day guide ships and planes even when GPS systems fail. All these innovations show how the smallest pieces of our universe can lead to some of the biggest advances in our future. Goldsmith explained how quantum computers could revolutionise fields like material science, [weather forecasting](#) and machine learning. **Daniel Goldsmith:** In the future, in fifteen, twenty years' time, fault-tolerant quantum computers will be developed. And at the moment, the problem with quantum computing is that the [qubits](#) are very [prone](#) to error. But when that problem is solved, then we'll be able to have powerful computations that are pretty much unimaginable now. We'll be able to have increases in material science. We'll be able to better understand the materials and the world that we live in, and build new advanced materials, maybe [room-temperature](#) semiconductors. We'll be able to potentially forecast the weather much better. Any computations that we do now will be able to be much better with the quantum computer. And [in the short term](#), we expect to see advances in

optimisation, in quantum machine learning, where the quantum computers will improve classical machine learning, and also in the simulation of physical systems, particularly quantum systems.

QUANTUM COMPUTING

Goldsmith then explained why building a quantum computer is so difficult, [outlining](#) the different approaches companies are taking, from superconducting circuits to trapped ions and photonic systems. **Daniel Goldsmith:** It's very difficult to build a quantum computer. There are all sorts of different modalities of quantum computers. There's lots of different ways of building the quantum computers. So for example, companies like IBM have [qubits](#) that are artificial atoms, and they're superconducting [qubits](#) on a chip. Other companies place ions, and they trap the ions on top of a chip and then control them by lasers. We've worked with Orca Computing, and they have photonic computers, where the computation is carried out by photons passing through [beam splitters](#). So there's lots of different ways of building the quantum computer. But what unites them all is it's extremely difficult to build a quantum computer, and it's even more difficult to get a quantum computer to do useful calculations.

AN OPTIMISED FUTURE

Finally, Goldsmith [addressed](#) the current state of the industry, noting that while commercial applications are limited today, businesses should prepare for a future where quantum computing transforms optimisation, simulation and AI. **Daniel Goldsmith:** Quantum computing is a [work in progress](#). There aren't really, at the moment, commercial opportunities for quantum computing. But we believe [it's well worth](#) companies understanding quantum computing, because in five, ten years' time, there'll be things that they can actually do in their business that will improve it. For example, optimisation, quantum machine learning, quantum simulation of physical systems. And in a decade, fifteen, twenty years' time, the quantum computing will be revolutionised. The most exciting thing is the way very abstract ideas, like entanglement, like superposition, like quantum phase,

actually produce really concrete results in terms of enabling computations that you simply couldn't do by any other means. If you want to know more about this topic, read the article [Schrödinger's Cat Explained](#).

Glossary

- **smarter** = più intelligenti, più efficienti
- **trace** = traccia
- **prone** = incline
- **room-temperature** = a temperatura ambiente
- **in the short term** = in tempi brevi
- **outlining** = sottolineare
- **behaviour** = comportamento
- **building blocks** = componenti di base
- **weather forecasting** = previsioni meteo
- **beam splitters** = divisori di fascio
- **it's well worth** = valere la pena
- **work in progress** = lavoro in corso
- **waves** = onde
- **MRI machines** = macchine per la risonanza magnetica (Magnetic Resonance Imaging)
- **qubits** = bit quantistici
- **splits** = dividersi
- **shows its hand** = mostrare le proprie carte, rivelare le proprie intenzioni
- **addressed** = affrontare
- **lump** = porzione